Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 9552
CALIBRATION DATE: 27-Feb-19

Slocum Payload CTD CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -9.900713e-001
h =  1.455886e-001
i = -1.752349e-004
j =  3.349368e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC = -3.1928e-008
```

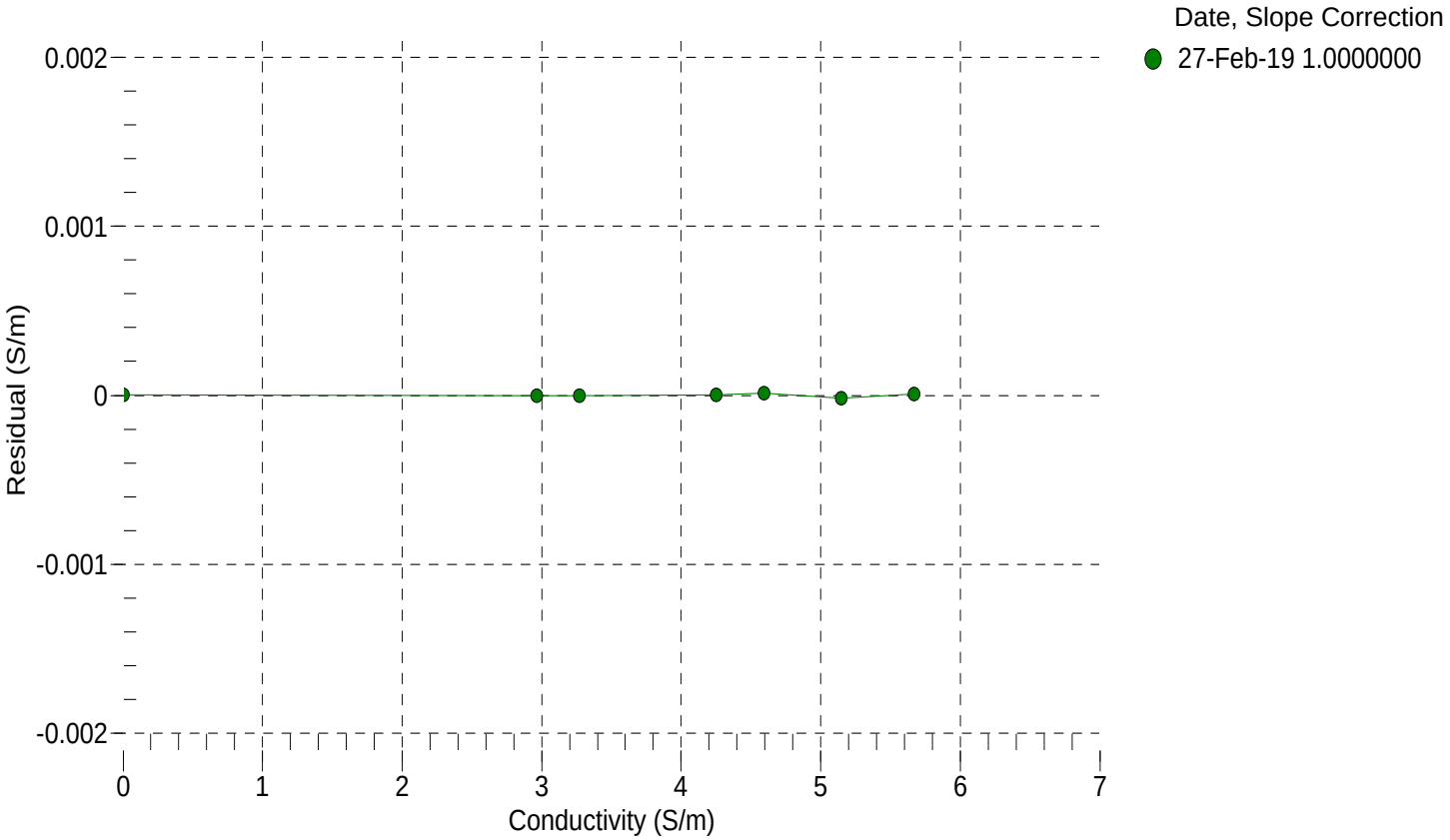
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2609.83	0.00000	0.00000
1.0000	34.6678	2.96448	5211.83	2.96448	-0.00000
4.5000	34.6483	3.27043	5409.05	3.27043	-0.00000
15.0000	34.6058	4.24851	5995.48	4.24852	0.00000
18.5000	34.5964	4.59233	6188.18	4.59235	0.00001
24.0000	34.5859	5.14813	6487.30	5.14811	-0.00002
29.0000	34.5789	5.66779	6754.74	5.66780	0.00001
32.5000	34.5733	6.03839	6939.01	6.03845	0.00006

f = Instrument Output(Hz) * sqrt(1.0 + WBOTC * t) / 1000.0

t = temperature (°C); p = pressure (decibars); δ = CTcor; ε = CPcor;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \varepsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 9552
CALIBRATION DATE: 18-Feb-19

Slocum Payload CTD PRESSURE CALIBRATION DATA
1450 psia S/N 11201506

COEFFICIENTS:

```
PA0 =      -1.085829e-001
PA1 =       4.560192e-003
PA2 =      -1.094738e-011
PTEMPA0 = -5.932953e+001
PTEMPA1 =  5.369333e-002
PTEMPA2 = -6.099097e-007
```

```
PTCA0 =  5.240752e+005
PTCA1 = -4.812242e+000
PTCA2 =  1.652478e-001
PTCB0 =  2.499513e+001
PTCB1 = -5.750000e-004
PTCB2 =  0.000000e+000
```

PRESSURE SPAN CALIBRATION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)
14.80	527309.2	1486.0	14.79	-0.00
301.06	590069.2	1497.0	301.07	0.00
588.08	652994.8	1498.8	588.01	-0.00
875.46	716057.5	1500.0	875.50	0.00
1162.77	779094.1	1501.6	1162.79	0.00
1449.89	842111.4	1502.8	1449.89	0.00
1162.88	779114.7	1503.6	1162.88	0.00
875.45	716029.5	1503.6	875.38	-0.01
588.29	653059.2	1505.2	588.31	0.00
300.98	590069.5	1507.0	301.07	0.01
14.80	527308.2	1509.7	14.78	-0.00

THERMAL CORRECTION

TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
32.50	1745	527362.80
29.00	1677	527341.40
24.00	1580	527325.40
18.50	1474	527303.20
15.00	1407	527314.20
4.50	1205	527327.20
1.00	1138	527336.80

TEMPERATURE (°C)	SPAN
-5.00	25.00
35.00	24.98

y = thermistor output (counts)

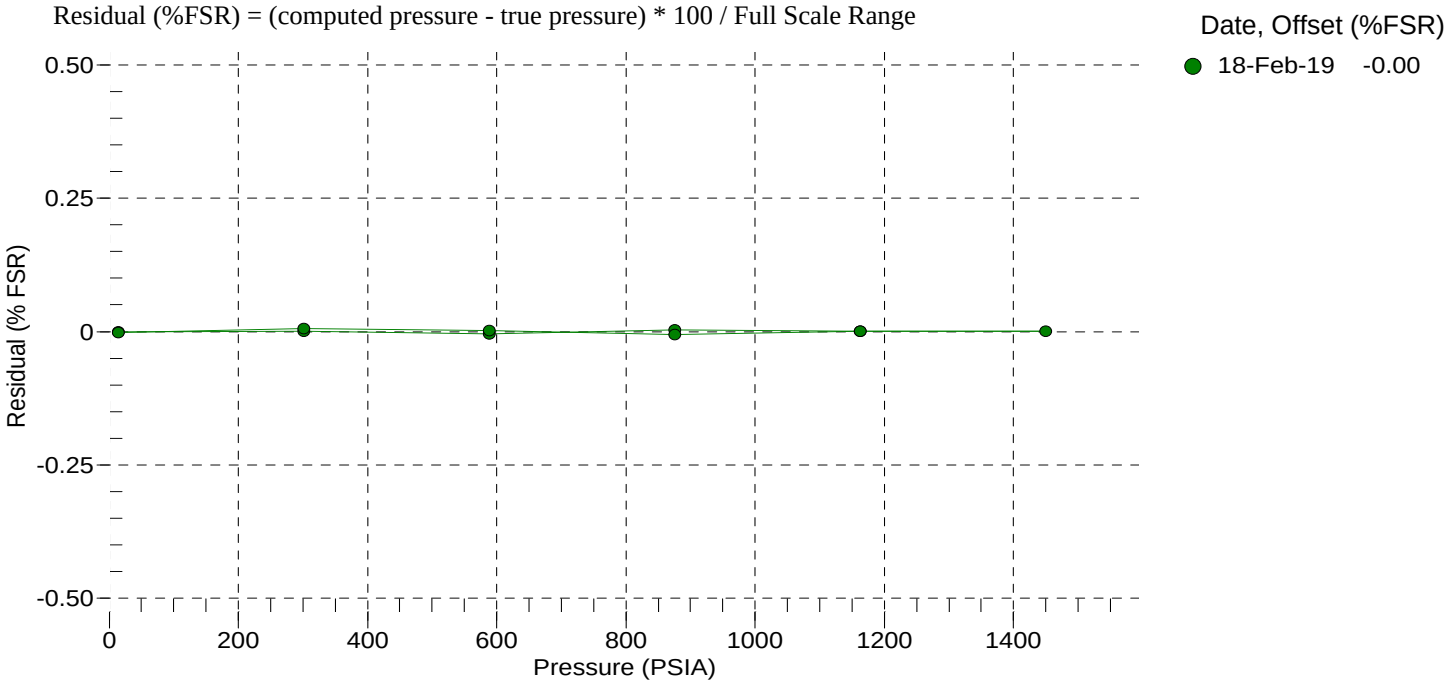
$t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range



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Pressure Test Certificate

Test Date: 2019-03-04

Description: Slocum CTD

Sensor Information:

Model Number: Slocum

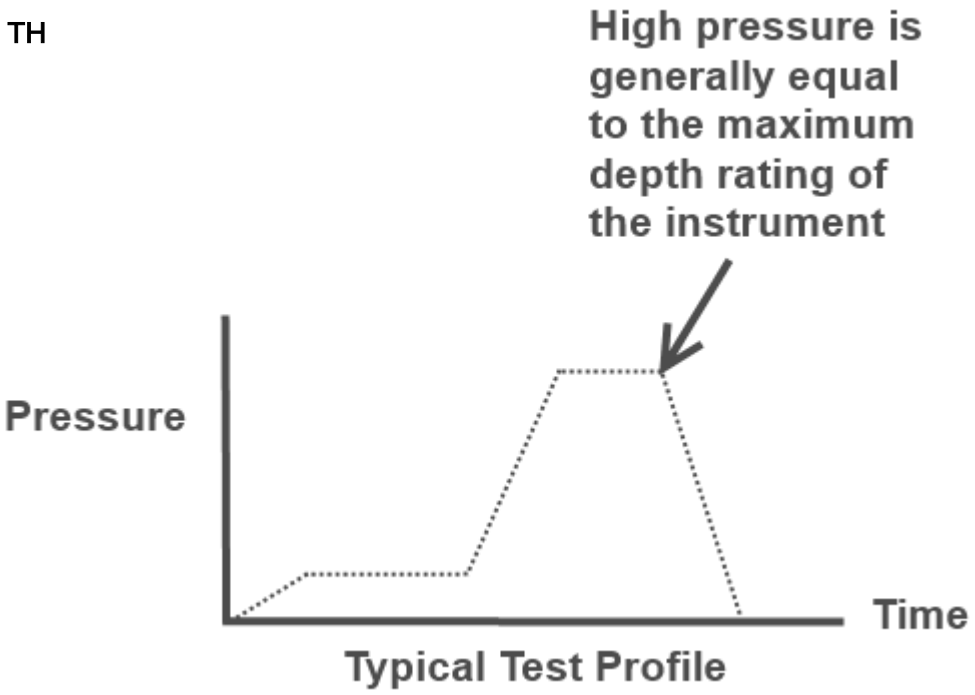
Serial Number: 9552

Pressure Test Protocol:

Low Pressure Test:	40	PSI	Held For:	15	Minutes
High Pressure Test:	1450	PSI	Held For:	15	Minutes

Passed Test: True

Tested By: TH



Typical Test Profile

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SENSOR SERIAL NUMBER: 9552
CALIBRATION DATE: 27-Feb-19

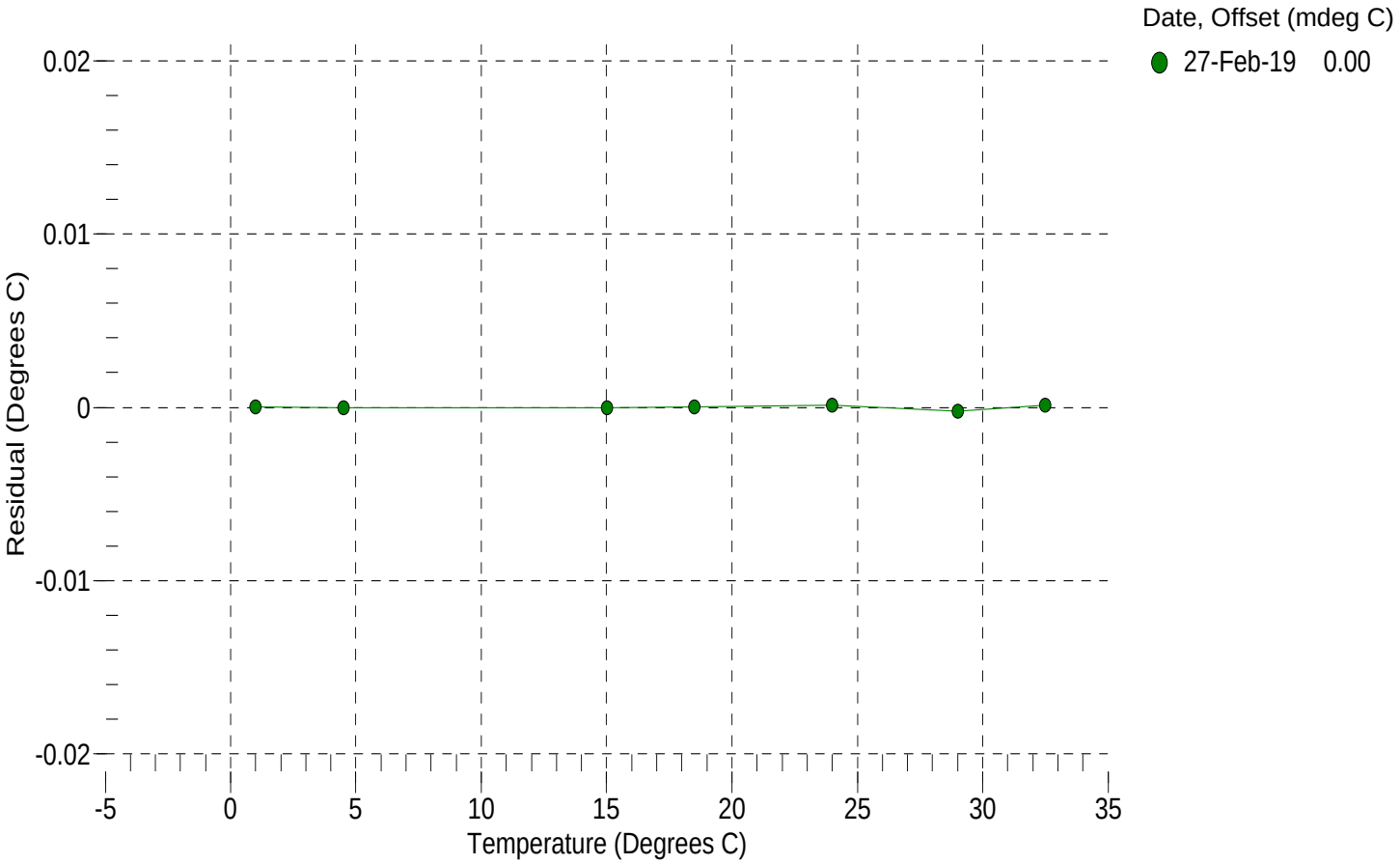
Slocum Payload CTD TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

```
a0 = -8.429746e-005
a1 =  3.000271e-004
a2 = -3.785648e-006
a3 =  1.818163e-007
```

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	564718.8	1.0000	0.0000
4.5000	483187.6	4.5000	-0.0000
15.0000	308688.0	15.0000	-0.0000
18.5000	267528.4	18.5000	0.0000
24.0000	214935.0	24.0001	0.0001
29.0000	177235.4	28.9998	-0.0002
32.5000	155367.6	32.5001	0.0001

n = Instrument Output (counts)

Temperature ITS-90 (°C) = 1/{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]} - 273.15

Residual (°C) = instrument temperature - bath temperature